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EVIDENCE FROM APRIL 2, 2025

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Rounding up the Effect of Tariffs on Financial Markets: Evidence from April 2, 2025
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ABSTRACT

We measure the response of financial outcomes to the US announcement on April 2, 2025, of tariffs on nearly all its trading partners. To address the challenge posed by potential anticipation by economic agents, we decompose these tariffs into a component associated with bilateral deficits and an arguably unanticipated component resulting from the rounding up of the continuous deficit measure to the next whole number or to the baseline 10% rate. We measure the short-term response of stock prices and exchange rates, focusing on all countries with which the US has a bilateral deficit. For both outcomes, the round-up component's effect is an order of magnitude larger than that of the deficit component. We find that a one percentage point higher tariff is associated with a statistically significant 0.23% decline in stock prices. Further, we find no evidence of a dollar appreciation; if anything, higher tariffs are associated with a dollar depreciation, driven by countries with a floating regime. We show this is consistent with a model that allows for trade reallocation and in which exports to the US are invoiced in dollars while exports to the rest of the world are partly invoiced in producer currency.

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1. INTRODUCTION

On April 2, 2025, in a historically unique policy action, the US announced tariffs on imports from nearly all trading partners. In the following hours, financial markets reacted across the globe. In principle, the variation in tariff rates across trading partners constitutes an unprecedented opportunity to measure the effect of these tariffs on financial outcomes. However, a key challenge is the potential anticipation of economic agents. In fact, two months earlier, a presidential memorandum had announced the study of “reciprocal” tariffs with the goal of reducing bilateral trade deficits, and the date for the April 2 tariff announcement had been set and publicized in advance. Thus, part of the impact on financial markets could have been absorbed before the announcement. We overcome this challenge by decomposing these tariffs into a component which is a function of bilateral deficits, which could have been potentially anticipated by markets, and an arguably unanticipated component due to the **rounding up** of the continuous deficit measure to the nearest next whole number or to the baseline 10% tariff rate.

We measure the short-term response of stock prices and exchange rates, focusing on all trading partners with which the US has a bilateral trade deficit. For stock prices, we use data on all listed firms in Compustat Global and find that the round-up component had a negative and statistically significant impact on stock market prices. The magnitude is such that a one percentage point higher tariff is associated with a 0.23% decline in stock prices. The impact of the potentially anticipated deficit component was also negative but an order of magnitude smaller, suggesting it had in fact been to a large extent anticipated. Strengthening these results, we see that this negative impact of the round-up component was stronger for tradable-sector firms than for service-sector firms, in line with the fact that tariffs affect tradable-sector firms more directly than service-sector firms. For large countries, the US tariff announcement could be read as also implying future retaliation, given that several large countries have retaliated to US tariffs in the past. We find that our results are robust to excluding these large countries from the sample. In addition, the tariff announcement listed a small set of product exclusions. We also find the results hold after removing firms in those industries. Our findings imply that, in any context, it is critical to distinguish between the anticipated and unanticipated components of a policy when studying its effects on financial markets.

In the case of exchange rates, we again observe that the round-up component drives the results. Our initial results indicate that there is no evidence of an appreciation of the US dollar. Across several specifications with various controls, we find a bilateral depreciation of the dollar. This is driven by countries with a floating exchange rate regime, and is statistically significant only in some of our regressions. This goes against what we would expect from existing theory, which indicates that US tariffs would reduce demand for foreign currencies, leading to a dollar appreciation. If US imports are invoiced in US dollars, we would expect no or small response in the exchange rate.

We find that these results are robust to including in our regressions a component capturing the product exclusions in the tariff announcement. We also check that this result is robust to controlling for the structure of each country’s exports. Specifically, we control for the share of intermediate inputs and

the share of commodities in exports to the US, the share of exports to the US over total exports, and the share of dollar-invoiced exports in total exports. None of these controls change our initial results.

An important point to consider is that the April 2 announcement, due to its magnitude, could have shaped the perception of markets about the US economy. This, in turn, could have weakened the US dollar. This effect is absorbed by the constant in our regressions, which are based on variation across trading partners. However, it is possible that countries differ in how much their currency responds to news about the US economy. In particular, countries with stronger financial linkages to the US could respond more to these news. If there happens to be a correlation between the level of April 2 tariffs received by different countries, and the strength of their financial linkages to the US, this could confound the interpretation of our results. We augment our regression to control for a measure of financial linkages to the US, and for capital account openness. However, results go in the same direction and under no scenario we find evidence of a dollar appreciation.

To make sense of this surprising response of the exchange rate, we provide a simple model that departs from the standard model of trade and exchange rates by adopting more realistic assumptions about invoicing currencies and about trade reallocation in response to tariffs. In line with recent empirical findings [Boz et al., 2022], we assume that countries facing US tariffs export to the US invoicing in dollars, and they export to the rest of the world in a combination of their own currency (i.e., producer currency) and US dollars. We further assume that US tariffs cause a decline in exports to the US but a reallocation of exports toward the rest of the world. This assumption is microfounded by assuming an increasing marginal cost and is consistent with recent evidence on the response to trade war tariffs [Benguria and Saffie, 2024]. Under these assumptions, a US tariff increases the relative demand for the currency of the country facing the tariff, leading to a bilateral dollar depreciation, in line with our empirical finding. The model also predicts a decline in stock prices, also in line with the evidence we provide.

A growing literature has documented the effects of tariffs spurred by the changes to US and global trade policy since 2018 [Amiti et al., 2019, Fajgelbaum et al., 2020, Flaaen et al., 2020, Cavallo et al., 2021, Benguria et al., 2022, Benguria and Saffie, 2024, Blanchard et al., 2024, Bonadio et al., 2024, Chor and Li, 2024, Fajgelbaum et al., 2024, Freund et al., 2024, Hankins et al., 2024, Benguria and Saffie, 2025, Gopinath et al., 2025, Handley et al., 2025, Flaaen and Pierce, forthcoming].

Recent work has begun to analyze the consequences of changes to US trade policy in 2025, including the April 2 tariffs we focus on in this paper. Ignatenko et al. [2025] develop a quantitative model to analyze the long-run effects of these tariffs and the possible retaliatory response of US trading partners. Rodríguez-Clare et al. [2025] provide an analysis of recent US tariffs on Canada, Mexico, China and of April 2 tariffs using a model with downward nominal wage rigidity and allocating the impacts across US states and foreign countries. Cavallo et al. [2025] measure the response of retail prices to the various trade policy actions of 2025.

There is also recent work that focuses on the link between trade policy and stock prices, in the context of the US-China trade war [Amiti et al., 2024, Huang et al., 2023] or previous episodes of trade

liberalization [Greenland et al. \[2024\]](#). Our contribution is to use the unique April 2 setting to isolate the unanticipated component of the tariff announcement, and to show that financial markets were quite effective at anticipating a large fraction (the deficit component) of this policy.

Moreover, there has been much progress recently in understanding the macroeconomic effects of trade policy [[Barattieri et al., 2021](#), [Jeanne, 2021](#), [Bergin and Corsetti, 2023](#), [Auray et al., 2024](#), [Aguiar et al., 2025](#), [Auclert et al., 2025](#), [Auray et al., 2025](#), [Bergin and Corsetti, 2025](#), [Bianchi and Coulibaly, 2025](#), [Cuba-Borda et al., 2025](#), [Monacelli, 2025](#), [Kalemli-Ozcan et al., 2025](#), [Werning et al., 2025](#)]. We believe our estimate of the response of the exchange rate to tariffs is an important input to these models. In fact, [Itskhoki and Mukhin \[2025\]](#) show that whether and to what extent the currency of a country imposing a tariff appreciates is key to understanding whether the tariff can close a trade deficit.

For good reason, we do not find much evidence of the elasticity of exchange rates to tariffs in the literature. Providing an exact elasticity ideally requires variation in tariffs across several countries imposed by a large player in the global economy. We believe April 2, 2025 is the first such event in history. [Jeanne and Son \[2024\]](#) study the response of the dollar/renminbi exchange rate to tariff news during 2018/2019. They find that the renminbi depreciates against the dollar in response to announcements of US tariffs. Similarly, [Matveev and Ruge-Murcia \[2024\]](#) document that US announcements of tariffs on Mexico and Canada lead to a depreciation of the Canadian dollar and Mexican peso against the US dollar. However, in both these papers the empirical setting does not allow them to establish an elasticity of exchange rates to tariffs or to isolate the unanticipated component of these policy announcements. Thus, our findings are not necessarily contradictory to these earlier results, because in their setting, the response of the exchange rate depends on how tariff announcements compare to market expectations.

2. CONTEXT AND DATA

2.1. Context

On January 20, 2025, the White House posted a memorandum entitled “America First Trade Policy” that set out the general principles for the trade policy of the entering administration. Section 2 in this memorandum addresses deficits and tasks the Secretary of Commerce with recommending “appropriate measures, such as a global supplemental tariff.”

On February 13, 2025, the White House posted a memorandum entitled “Reciprocal Trade and Tariffs”. The memorandum argues that the US has been very open to trade, and describes a lack of reciprocity in the treatment it receives from its trading partners. The memorandum also argues that this lack of reciprocity has resulted in a US deficit in goods trade. The document then mentions a plan to address this which will include “reciprocal tariffs”. It also states these tariffs will be a function of an assessment of the fairness and balance of trade with each country. Finally, the memorandum emphasizes that a report will be produced describing the precise measures applied to each trading partner.

During an address to a joint session of Congress on March 4, 2025, the US president set the date for reciprocal tariffs for April 2. There was much discussion in the press about these tariffs between then

and the day of the announcement.

April 2 tariffs were announced at the White House shortly after 4:00 p.m. Eastern Time (E.T.). Further information was posted that evening by the White House in an Executive Order entitled “Regulating Imports with a Reciprocal Tariff to Rectify Trade Practices that Contribute to Large and Persistent Annual United States Goods Trade Deficits”, including all the details of the policy.

By the time the announcement began, the US stock market had just closed. The stock markets in all the countries in our sample were also closed.¹ Our analysis is based on comparing closing prices on April 2 (i.e. pre-announcement) to closing prices on April 3. Our sample consists of countries with which the US had bilateral deficits, mostly in Asia, Africa and Europe, and a few in the Americas.

2.2. The tariff formula

The formula used to calculate April 2 tariffs is described in the document entitled “Reciprocal Tariff Calculations” published by the US Trade Representative (USTR).² Define US imports from country i as m_i and US exports to country i as x_i . The document defines a “reciprocal tariff” toward country i based on the formula:

$$\Delta\tau_i = \frac{1}{\epsilon \cdot \varphi} \cdot \frac{x_i - m_i}{m_i}. \quad (1)$$

The authors argue that this is the additional tariff that would be needed to close the bilateral trade deficit. In this expression, ϵ is the price elasticity of import demand and φ is the elasticity of import prices with respect to tariffs. The authors of the document chose parameter values $\epsilon = 4$ and $\varphi = 0.25$, such that these terms cancel out. For our purposes, the only relevant point is that these “reciprocal” tariffs are a function of bilateral trade deficits.

The tariffs announced on April 2 correspond to dividing these values by two and, key to our endeavor, rounding them up to the next whole number. Throughout the paper, we refer to the unrounded value as the “deficit component” of the tariff. We refer to the rounding up part of it as the “round-up component”. This means that:

$$\text{Total tariff}_i = \text{Deficit component}_i + \text{Round-up component}_i \quad (2)$$

Countries with which the US has a trade deficit and for which the reciprocal tariff divided by two is less than 10% receive a baseline 10% tariff. In other words, there is also a round-up component for these countries.

For example, the US has a bilateral trade deficit with Norway (measured as exports minus imports over imports) of -30.248%. This leads to a deficit component of the tariff of 15.124%. The announced tariff is 16%. This means that the round-up component is 0.876%.

Countries with which the US has a trade surplus also receive the baseline 10% tariff. However, our

¹The exception might be Ecuador, which according to regular trading hours might have closed 15 minutes after the announcement. We show later our results are robust to excluding it.

²This document was published more than a week later than the April 2 announcement, so it is not part of the information agents had during our study window.

analysis excludes them as we focus on countries with which the US has a trade deficit. Our reason to exclude countries with which the US has a bilateral surplus is the following. For these countries, it is not clear that we can distinguish between a “trade balance” (in this case surplus) component and a round-up component, as tariffs have a zero lower bound. In other words, we believe the experiment is much cleaner and well-defined when focusing on deficit countries and the goal of our paper is to use the nature of the tariff formula to provide credible estimates of the response of stock prices and exchange rates to tariffs.

A few countries were not assigned April 2 tariffs. These include Mexico and Canada, given that the US had earlier separately imposed large tariffs on these countries. It also includes Russia, Cuba, Belarus and North Korea in light of existing sanctions.

For other (typically very small) countries, the US deficit would have implied that they would be listed with tariffs beyond the 10% baseline rate, but they were not included in Annex I, thus facing the 10% rate.³ These countries are excluded from our analysis.

The April 2 executive order excludes a small set of products including copper, pharmaceuticals, semiconductors, etc. It provides a list of excluded HS8 codes in “Annex II”. We explain below how we account for these product exclusions in our empirical analysis.

2.3. Data

Trade flows and tariffs We obtain the list of April 2 tariffs from the Executive Order published by the White House immediately after the announcement.⁴ Data on US exports and imports vis-a-vis each trading partner in 2024 comes from the Census Bureau.⁵ The note on “Reciprocal Tariff Calculations” by USTR provides summary statistics on the unrounded deficit component of the April 2 tariffs. We verify that the summary statistics in the measures we compute match those.

Our analysis is focused on countries with which the US had a bilateral trade deficit in 2024. There are 82 countries or territories with which the US has a trade deficit. We exclude 12 countries on which the US did not impose April 2 tariffs. Out of the remaining countries, 29 are in Asia/Oceania, 25 in Africa, and 7 in the Americas. The European Union was treated by the announcement as a single entity, so we also treat it as such. There are 8 European (extra-EU) countries in our sample.

Most of the countries in our sample appear in Annex I of the April 2 announcement, which lists the countries receiving a tariff beyond the 10% rate given that the US has a large trade deficit with them. A handful of countries with which the US had a smaller deficit such that they were imposed the baseline 10% rate do not appear in Annex I but are also included in our analysis.

³This group includes the following countries and territories: Faroe Islands, Saint Pierre and Miquelon, Niue, Reunion, Norfolk Island, Afghanistan, and the West Bank. This group might include Bolivia, which should have received an 11% tariff according to the formula but received the 10% baseline rate (or an alternative view is that Bolivia was slightly rounded down). In any case, we also exclude Bolivia from the sample.

⁴Links to this and other relevant documents are available in the Appendix.

⁵Specifically, the data comes from the Census Bureau’s website on U.S. Trade in Goods by Country. Note that the note on “Reciprocal Tariff Calculations” published by USTR confirms that data from the Census Bureau for 2024 was used in the calculations.

Table 1 includes the list of all countries with which the US has a trade deficit. It reports bilateral US imports and exports, and the trade deficit defined as $\frac{\text{imports} - \text{exports}}{\text{imports}}$ (as defined in the USTR note). It also reports the total April 2 tariff announced, the deficit component, and the round-up component.

Stock prices We use the universe of listed firms included in Compustat Global. We focus on the difference between the closing price in April 2 and April 3. We exclude from the main sample firms missing an industry code, although in robustness checks we find this does not affect the results. We focus on firms located and incorporated in the same country, reporting stock prices in local currency.⁶ For firms issuing more than one security, we consider the stock price of the primary one. Appendix Table 2 shows the set of countries available in Compustat Global and the number of listed firms in each.

Exchange rates We obtain exchange rates from a Bloomberg terminal. This has the advantage that it has broad coverage in terms of countries available. It also has the benefit of reporting intraday exchange rates, which allows us to study their response at different horizons (24, 12 and 3 hours). We express all exchange rates in local currency per US dollar. We exclude from the sample countries that do not have their own currency, are pegged to another currency, or (in very few cases) for which exchange rate data is not available. This leaves 40 countries in our sample (not all available for all the horizons). We obtain exchange rate regimes from the IMF's Annual Report on Exchange Arrangements and Exchange Restrictions (AREAER) of 2023. There are 17 countries with a floating regime. Appendix Table 3 reports the set of countries in the sample used to estimate the impact on exchange rates, and the countries excluded and the reason for it.

For context, Figure 1 plots the dollar index at high frequency between April 1 and April 3. It shows a 1.7% depreciation of the US dollar over the 24 hours following the tariff announcement.

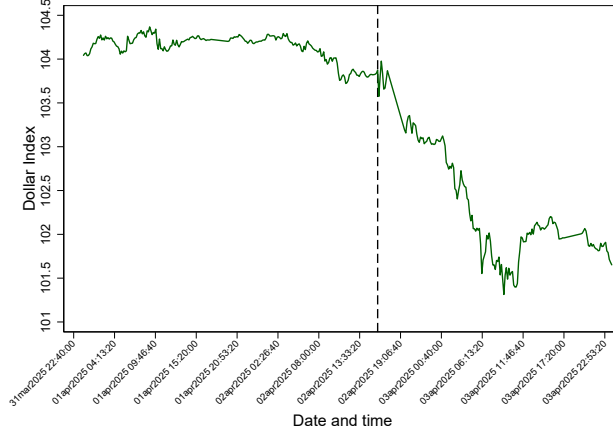
Excluded products Annex II in the April 2 executive order provides a list of 1039 excluded HS 8-digit codes. For comparison, the US imported 10,908 HS 8-digit codes in 2024. Because not all products are imported from all countries, excluded products account for 6.5% of all the product–country combinations. The list of excluded products accounts for 19.4% of the imported value in 2024. However, note that the value of excluded products is extremely concentrated among few products. In fact, six HS 4-digit industries account for 75% of this value. Among these, some oil and pharmaceutical products account for the largest share.

In the firm-level analysis for stock prices, we know the NAICS industry of each firm in Compustat Global, and can exclude firms in industries with a significant share of excluded products. For this purpose, we compute the share of excluded products by NAICS 2-digit industry and by trading partner, weighted by US imports in 2024.⁷ Specifically, we start with data from the Census Bureau on US imports

⁶For the countries in our sample, 99.3% of firms report stock prices in local currency. Regarding firms with stock prices reported in currencies different than the local currency, we choose not to transform them to local currency because we do not know what value was used for the exchange rate, and we do not want the response of the exchange rate to confound the stock price results.

⁷Note that most often firms in Compustat Global do not have industry codes at very disaggregate levels.

Figure 1: Dollar index



Notes: This figure plots the dollar index during April 1 - April 3, 2025. The dashed vertical line marks 4:00 p.m. Eastern Time on April 2. The dollar index is obtained from Bloomberg and is defined as the value of the US dollar relative to a basket of six currencies (euro, Japanese yen, British pound, Canadian dollar, Swedish krona, Swiss franc). An increase in the dollar index corresponds to an appreciation of the dollar.

by HS 10-digit code and trading partner for 2024. We merge in NAICS industries using a concordance from the Census Bureau between HS 10-digit codes and NAICS 6-digit industries. We also merge in the list of excluded products at the HS 8-digit level. We then construct the value-weighted share of excluded products by NAICS 2-digit industry and trading partner as described earlier.

In the country-level analysis for exchange rates, we take a different approach. For each trading partner, we compute an average tariff that excludes the products in Annex II (assigning them a zero tariff). This is a weighted average, with weights equal to US imports in 2024. We then let decompose the overall tariff announced into three components: the deficit component, the round-up component and a component equal to the difference between total tariff announced and the weighted tariff accounting for exclusions.

For example, for Thailand the total tariff announced is 37%. The deficit component of the tariff is 36.01% and the round-up component is 0.99%. Assigning a zero tariff to excluded products and weighting by US imports in 2024, the weighted tariff for Thailand is 35.46%. That means the exclusions component is $35.46\% - 37\% = -1.54\%$. Thus, we can write the weighted tariff of 35.46% as the deficit component, plus the round-up component, plus the exclusions component ($36.01\% + 0.99\% - 1.54\%$).

Table 4 reports the component associated with product exclusions for each country in the sample used for the exchange rate regressions.

Additional data Other datasets used as control variables in some regressions and measuring the structure of countries' exports as well as financial characteristics are described in the Appendix.

3. RESULTS

We first present results for stock prices and then move on to exchange rates.

3.1. Stock prices

We estimate the following regression of log changes in stock prices on tariffs, decomposing the total tariff into the deficit component and round-up component as described earlier. Given that this policy impacts all firms, we do not measure a firm's returns relative to the market. The dependent variable is $(100 \times)$ the change in the stock price of firm f measured as the log difference in the closing price between April 2 and April 3.

$$\Delta Y_f = \beta_1 \cdot \text{Deficit component}_c + \beta_2 \cdot \text{Round-up component}_c + \gamma_i + \epsilon_f. \quad (3)$$

We estimate this equation by OLS. Given that the tariffs vary by country, we cluster standard errors by country. Because some countries have many more listed firms than others, we weigh observations by the inverse of the number of listed firms in each country. We control for industry fixed effects (γ_i) based on 2-digit NAICS industries.

The results are presented in Table 5. In column 1 we include only the total tariff instead of its components, finding a negative coefficient that is not statistically significant. The baseline results in column 2 indicate that the coefficient on the round-up component is negative and statistically significant. It is an order of magnitude larger than the coefficient on the deficit component, which is also statistically significant. The magnitude of the coefficient on the round-up component is such that a one percentage point higher tariff is associated with a 0.23% decline in stock prices. The average round-up tariff component imposed to the countries with which the US has a bilateral deficit listed in Table 1 is 1.14%. Using this as a benchmark and the elasticity just discussed would imply a 0.26% decline in stock prices. We can also use the 0.23% decline per percentage point to calculate that the 25.6% average total tariff would imply 5.9% decline in stock prices. This should not be compared to the observed decline in stock prices between April 2 and April 3. As our results show, much of the deficit component of the total tariff appears to have been anticipated, as the response is much lower compared to the round-up component. Therefore, one should compare that calculated 5.9% to the decline in stock prices over a longer period of time before the announcement.

It is possible that upon the announcement of US tariffs, agents assume that this would also imply retaliation by the countries facing the tariffs. Based on the response to the 2025 tariffs and past experience from the 2018-2019 period, it is reasonable to assume that large countries might retaliate but not smaller ones. In column 3, we report results excluding large countries. We classify as large countries those at least as large in terms of GDP as the smallest country that retaliated against the US in 2018-2019, which was Turkey.⁸ We see results in the same direction, with a larger coefficient on the round-up

⁸We consider China, the European Union, India, Indonesia, Japan, South Korea, and Turkey in the group of large countries.

component of -0.33. The coefficient on the deficit component is again much smaller.

In column 4 we exclude firms in industries facing product exclusions. As described in Section 2.3, we compute the share of imports by country and NAICS 2-digit industry that is excluded from tariffs. We then remove from the sample all firms in industry-country combinations in which more than 10% of the imported value falls under the product exclusions. Coefficients are quite similar. Further, in column 5 we both exclude large countries and remove firms facing product exclusions.

Next, we distinguish between firms in tradable (manufacturing, agriculture and mining) and non-tradable (service) industries. We would expect that the response of stock prices to tariffs is larger among firms in the tradable sector, as these firms are directly impacted by them. In columns 6 and 7 we see that the coefficients on the round-up component are -0.41 and -0.31 for the tradable and nontradable sectors respectively. The coefficient on the deficit component is also larger for the tradable sector, and again, much smaller than that for the round-up component.

We also go through a series of robustness checks, finding similar results across the board. These are shown in Table 6. In all columns we exclude large countries and firms facing product exclusions (like in column 4 of Table 5). In column 1 we remove industry fixed effects.⁹ In column 2 we replace the NAICS 2-digit by NAICS 3-digit industry fixed effects. In column 3, we winsorize changes in stock prices at the 1st and 99th percentiles. In column 4 we restrict the sample to countries with at least 10 listed firms. Finally, in column 5 we remove certain countries facing special situations. These are Ecuador because its stock market would have been open for a small period of time after the tariff announcement on April 2 based on its regular hours, and South Africa because of the macroeconomic volatility on April 3 due to the introduction of a budget law that caused political tension. Overall, we see little changes across all these specifications.

3.2. Exchange rates

We estimate a similar regression with $(100 \times)$ log changes in exchange rates as the dependent variable. However, this time we have one observation per country. For this reason, observations are not weighted, there are no fixed effects, and standard errors are not clustered. Recall that the exchange rates are defined in terms of local currency (i.e., currency of each trading partner) per US dollar.

$$\Delta Y_c = \beta_1 \cdot \text{Deficit component}_c + \beta_2 \cdot \text{Round-up component}_c + \epsilon_c . \quad (4)$$

The timing we use to measure exchange rate changes is as follows. In foreign exchange markets, there is not such a clear open and close time each day as in stock markets. For some exchange rates, we see values over the 24 hours each day. For smaller countries, data is available during certain windows each day. We measure the initial value for each exchange rate as the latest value available before 4:00 p.m. Eastern Time (ET) on April 2nd. For most countries, there is data at exactly 4:00 p.m. ET. We measure the final value at different horizons. We start with the change during a 24 hour period for our baseline

⁹This expands the number of observations as some firms are missing a NAICS industry code.

results. We later explore changes over 12 and 3 hour windows.

The results are shown in Table 7. In column 1 we include only the total tariff instead of its components, finding a coefficient that is not statistically significant. The baseline results in column 2 indicate that the coefficient on the round-up component is negative and an order of magnitude larger than the coefficient on the deficit component, which is closer to zero. Both coefficients are not statistically significant. The negative coefficient on the round-up component implies an depreciation of the US dollar (less local currency per US dollar). In column 3 we exclude large countries (that might retaliate), while in columns 4 and 5 we switch to a 12-hour and 3-hour window respectively, again with coefficients that are not statistically significant. Note that the number of observations falls substantially with the shorter windows, reason why we consider the 24-hour window our preferred specification.

The results presented so far pool countries with floating and non-floating exchange rate regimes together. Logically, one would expect more clear evidence from floaters, for which the exchange rate can react freely. There are 17 countries with a floating regime in our sample. For this reason, we do not split the sample but instead include an interaction term between each tariff component and a dummy variable for floaters:

$$\begin{aligned}\Delta Y_c = & \beta_1 \cdot \text{Deficit component}_c + \beta_2 \cdot \text{Round-up component}_c \\ & + \beta_3 \cdot \text{Floating}_c \cdot \text{Deficit component}_c + \beta_4 \cdot \text{Floating}_c \cdot \text{Round-up component}_c \\ & + \beta_5 \cdot \text{Floating}_c + \epsilon_c.\end{aligned}\tag{5}$$

These results are shown in Table 8. When using the total tariff (in column 1), coefficients are small, not statistically significant, and there is no clear difference between floaters and non-floaters. In columns 2 through 5, we distinguish between the round-up and deficit components. Columns 2 and 3 (which excludes large countries) consider 24-hour windows, while columns 4 and 5 move to 12-hour and 3-hour windows. Across these four columns we find a negative coefficient on the interaction between the round-up component and the dummy for countries with floating regimes. Also in all cases, the total effect for floaters is negative, implying a dollar depreciation. The coefficient in column 2 is such that a one percentage point higher tariff is associated with a 0.25% dollar depreciation relative to non-floaters. The coefficient on the interaction term is statistically significant only with the shorter windows.

Given that our sample has a low number of observations, we check that this result is not driven by any outlier. Dropping one country of the sample at a time, the coefficient on the interaction term between the round-up component and the dummy for floaters in the specification corresponding to column 2 is always negative (and not statistically significant) and moves between -0.166 and -0.604.

Excluded products Because the exchange rate regressions are based on one observation for countries, the approach to deal with excluded products is different than in the stock price regressions. Instead, we compute an average country-level tariff that takes into account product exclusions. As discussed in Section 2.3, this assigns a zero tariff to excluded products and weighs products by US imported value in 2024.

We then define the “exclusions component” of the tariff as the gap between the total tariff (the April 2 tariff announced for each country) and the lower weighted tariff accounting for exclusions. This implies that:

$$\text{Total tariff}_i = \text{Deficit component}_i + \text{Round-up component}_i + \text{Exclusions component}_i. \quad (6)$$

such that the exclusion component is always negative. We then estimate new versions of equations (4) and (5) including these three components on the right hand side.

These results are shown in Table 9. Columns 1 through 4 consider a 24-hour window, while columns 5 and 6 consider a 12-hour window. In column 1 we include only the weighted tariff that accounts for exclusions, with results that are small and not statistically significant. In column 2 includes the three components separately, and again find that the round-up component is again negative, and an order of magnitude larger than the deficit component and the exclusions component. Like before, the round-up component is not statistically significant.

In columns 4 through 6 we include interactions with the dummy for countries with a floating exchange rate regime. For the weighted tariff accounting for exclusions (see column 4), coefficients are again small and not statistically significant. In column 5, we have the results with the separate components and their interaction with the dummy for a floating regime. Here, like before, we find a negative (albeit not statistically significant) coefficient for the interaction between the round-up component and the dummy for floaters. Remarkably, the magnitude of the interaction term is very close to the one found in column 2 in Table 8. In contrast, the coefficients corresponding to the interaction terms with the other components are an order of magnitude smaller and not statistically significant. Results are similar in columns 5 and 6 considering a 12-hour window. In this case, the coefficient on the interaction term between the round-up component and the dummy for floaters is larger and statistically significant.

Overall, we conclude that the results for the round-up component are robust to accounting for product exclusions. In addition, we find that markets did not respond much to variation in tariffs coming from excluded products. One interpretation is that it was easier for investors to process the information on the much publicized total tariff than going over the list of HS codes in Annex II. Alternatively, investors could have also anticipated the product exclusions. There might have been some ground for this. For example, copper products are among those excluded, and could be related to the investigation on US copper imports announced a few weeks before April 2.

Controlling for trade structure and financial linkages to the US So far, we have lack of evidence in favor of a dollar appreciation, and if anything, evidence of a dollar depreciation. Evidently, this goes against what we would expect from theory. Here, we inspect our results further to see if we find an explanation for this discrepancy.

First, we control for the type of goods exported by countries, and the currency in which they are invoiced. It could be that higher tariffs happened to be placed on countries for which their trade structure involved a smaller response of their exchange rate. Specifically, we control for the share of exports to the US that are product inputs, and for the share of exports to the US that are commodities. Further, we

control for the share of total exports that go to the US. Finally, we control for the share of total exports invoiced in US dollars. The results are shown in columns 1 through 4 in Table 10. All columns consider a 24-hour window. Regardless of whether these control variables are introduced separately (columns 1 through 3) or all at the same time (column 4), results do not change much.

A second point is that the April 2 tariff announcement was an important event that might have changed perceptions about US policy and the US economy (or even the world economy as a whole). Given the center role of the US in the world, this could have led to a response of exchange rates beyond the standard channel of tariffs on exchange rates, triggering movements of capital flows toward or away from the US. Further, this response to meaningful news about the US economy can be heterogeneous across countries, and presumably stronger for countries with stronger financial linkages to the US, or with a more open capital account. If April 2 tariff levels happen to be correlated with elements that shape the response of the exchange rate to large news about the US economy, this could explain the sign of the coefficient we observe. To this end, we control for capital account openness captured by the Chinn-Ito index and by the share of portfolio liabilities of each country vis-a-vis the US over the country's total portfolio liabilities.¹⁰ The results are shown in columns 5 through 7 in Table 10. Once again, however, results do not change much, as the coefficient on the interaction term between the round-up component and the dummy variable for floaters is still negative.

Further, in column 8 in the same table we combine all controls - those capturing countries' trade structure and those capturing financial linkages to the US. Results point in the same direction, and the interaction term even becomes statistically significant. Thus, we conclude that there is no evidence of an appreciation of the US dollar. Depending on how one interprets these results, and keeping in mind the small number of observations, we could say that if anything, there are signs of a dollar depreciation.

4. A MODEL OF TARIFFS, EXCHANGE RATES, AND TRADE REALLOCATION

In the standard model of trade and exchange rates, a US tariff would reduce demand for foreign currencies, leading to a dollar appreciation. We develop a simple model which departs from the standard model by changing its assumptions about invoicing currencies and about trade reallocation in response to tariffs. In both regards, our assumptions are guided by well-established empirical facts. This simple variation delivers the opposite result: a US tariff leads to a dollar depreciation.

The assumption about invoicing patterns is the following. Exports to the US are invoiced in US dollars. Exports to the rest of the world are invoiced partly in dollars and partly in the currency of the exporting country (producer currency, using the terminology in the literature).¹¹ This assumption is consistent with the evidence in the literature on invoice currencies in trade [Boz et al., 2022, Gopinath and Rigobon, 2008].

¹⁰Financial linkages to the US can be measured in different ways, and our choice is driven by data availability, which is limited for small and underdeveloped countries in our sample.

¹¹In fact, it would be enough to assume that the share of exports invoiced in dollars is larger for exports to the US than for exports to the rest of the world.

The assumption about export reallocation is that US tariffs lead to a decline in exports to the US but an increase in exports to the rest of the world. This reallocation can be microfounded by assuming an increasing marginal cost. This mechanism is featured in [Benguria and Saffie \[2024\]](#) and [Almunia et al. \[2021\]](#). Further, [Benguria and Saffie \[2024\]](#) show that there is a substantial amount of trade reallocation in response to trade war tariffs.

Combining both assumptions, a tariff imposed by the US on a given country leads to a decrease in exports to the US and an increase in exports to the rest of the world. Because only exports to the rest of the world are invoiced in producer currency, this leads to a higher relative demand for producer currency. This implies an appreciation of the producer currency (i.e., a depreciation of the US dollar).

We now provide a mathematical overview. In Appendix Section A.4, we provide the microfoundations for this model by assuming an increasing marginal cost. For this purpose, we extend the model in [Benguria and Saffie \[2024\]](#) by including assumptions about invoice currencies.

The structure is the following. Consider a small economy that exports to the US and to the rest of the world (ROW). Following the microfoundation provided in the Appendix, exports to a given destination (US or ROW) are a function of both US and ROW tariffs. A US tariff leads to a decline in the value exported by this country to the US, and an increase in the value exported to ROW.

Exports to the US are:

$$r_{US} = A_{US} \cdot f(\tau_{US}, \tau_{ROW}). \quad (7)$$

In this expression, A_{US} captures the demand level in the US, and τ_{US} and τ_{ROW} stand for US and ROW tariffs imposed on this exporting country.¹² As shown in the Appendix, $\frac{dr_{US}}{d\tau_{US}} < 0$.

Exports to ROW are:

$$r_{ROW} = A_{ROW} \cdot f(\tau_{US}, \tau_{ROW}). \quad (8)$$

with $\frac{dr_{ROW}}{d\tau_{US}} > 0$. This means that a US tariff results in an increase in exports to the rest of the world, as there is export reallocation.

All exports to the US are invoiced in dollars. A fraction ϕ of exports to ROW is invoiced in dollars, while the remaining fraction $1 - \phi$ is invoiced in producer currency. Consequently, exports invoiced in dollars are $r_{US} + \phi \cdot r_{ROW}$ and exports invoiced in producer currency are $(1 - \phi) \cdot r_{ROW}$.

The exchange rate e between this exporting country's currency (i.e., producer currency) and the US dollar is given by the ratio between exports in producer currency and exports in dollars.¹³ That is:

$$e = \frac{(1 - \phi) \cdot r_{ROW}}{r_{US} + \phi \cdot r_{ROW}}. \quad (9)$$

As discussed earlier, a US tariff leads to an increase in r_{ROW} and a decrease in r_{US} . This means that the exchange rate e increases, such that the producer currency appreciates relative to the US dollar.¹⁴

While it is not essential, one can think that the demand for dollars will not change as the exporting

¹²For simplicity, one could disregard the ROW tariff without loss of generality.

¹³This means that the exchange rate is demand-determined and we assume the supply of each currency is fixed.

¹⁴A mathematical proof is provided in the Appendix.

country is small relative to the world. Thus, the denominator does not change, and again a US tariff leads to an appreciation of the producer currency.

Endogenous currency choice We can extend the model to a case in which the invoice currency shares respond to a US tariff. Recent theories model invoice currencies as endogenous [Amiti et al., 2022]. Further, while much evidence shows that invoice currencies are quite stable, recent evidence shows that large and fast changes in invoice currencies are possible [Benguria and Novy, 2025, Crowley et al., 2024]. Specifically, we assume that in response to a US tariff, exports to the US remain invoiced in dollars and exports to the rest of the world shift toward a higher share of invoicing in producer currency (i.e., invoicing moves away from the dollar). In other words, the parameter $\phi = \phi(\tau_{US})$ is endogenous and $\frac{d\phi}{d\tau_{US}} < 0$. In the Appendix, we show that in this scenario the appreciation of local currency in response to a US tariff is stronger than in the baseline case.

Stock prices This model also predicts a decline in stock prices in response to a US tariff. In the Appendix, we compute the effect of a tariff on the total profits of the representative firm. We assume stock prices depend on the present discounted value of future profits, such that they fall as a consequence of a US tariff.

5. CONCLUSIONS

We use a first-in-history natural experiment, the April 2 2025 tariffs imposed by the US at varying rates on all trading partners, to measure the effect of tariffs on financial markets. We overcome the key challenge of these effects being anticipated by markets by exploiting a feature of the tariff formula used to set these tariffs. The tariffs were a function of US bilateral trade deficits with each country, but were rounded up to the next whole number or to the baseline 10% tariff rate. The round-up component of the tariffs is arguably entirely unanticipated.

The results indicate that indeed the round-up component is associated with a response that is an order of magnitude larger than the potentially anticipated deficit component. We measure a negative and statistically significant impact of tariffs on stock prices, such that a one percentage point higher tariff (as captured by the round-up component) is associated with a 0.23% decline in stock prices. Reassuringly, this effect is larger for firms in tradable industries than for firms in the service sector. Our results hold after excluding large countries, which might retaliate, and after accounting for product exclusions in the policy.

We also measure the response of exchange rates to tariffs. This is a key elasticity to inform macroeconomic models designed to understand the optimal policy response to tariffs. Again, the impact on exchange rates is driven by the round-up component. Once we distinguish between those countries with a floating exchange rate regime and the rest, our initial results indicate that there is no evidence that higher tariffs are associated with an appreciation of the US dollar relative to the currency of the countries facing these tariffs. The coefficients we find across several specifications go in the opposite

direction, and are statistically significant in some cases. This goes against what we would expect from international macroeconomic models, which would suggest a dollar appreciation (small if most US imports are invoiced in dollars). These initial results are robust to controlling for the trade structure of each country’s exports, including the shares of intermediate inputs and of commodities in exports to the US, the share of exports to the US in total exports, and the share of total exports invoiced in dollars. Further, the April 2 announcement might have been read as a large shock changing perceptions about the US or global economies. If exchange rate responses to this type of news are heterogeneous across countries and happen to be correlated with tariffs, this could potentially explain our findings. However, controls addressing this explanation do not alter our results.

To reconcile our finding on the effect of US tariffs on exchange rates, we present a simple model that departs from standard theory in two dimensions: allowing for trade reallocation and assuming exports to the US are invoiced in dollars while exports to the rest of the world are split between dollars and producer currencies. These assumptions are consistent with well-established empirical facts, and in this scenario a US tariff leads to a dollar depreciation.

To conclude, we emphasize that the large gap in the estimated coefficients between the potentially anticipated deficit component and the arguably unanticipated round-up component suggests that anticipation effects in financial markets are important. This implies that any existing or future estimates that we have on the effect of policies on financial outcomes should be assessed carefully in a context-dependent way in light of potential anticipation by markets.

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Table 1: List of trading partners with which the US has a bilateral trade deficit (**Part 1**)

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Country Name	US imports	US exports	US deficit	Total tariff	Deficit tariff component	Round-up tariff component
Lesotho	237.3	2.8	98.822	50	49.411	0.589
Cambodia	12661.8	321.6	97.460	49	48.730	0.270
Laos	803.3	40.4	94.973	48	47.486	0.514
Madagascar	733.2	53.4	92.716	47	46.358	0.642
Vietnam	136561.2	13098.2	90.409	46	45.204	0.796
Myanmar	656.5	77.2	88.247	45	44.124	0.876
Sri Lanka	3015.6	368.2	87.790	44	43.895	0.105
Falkland Islands	22.8	4.1	82.016	42	41.008	0.992
Syria	10.7	2.0	81.028	41	40.514	0.486
Mauritius	234.5	48.0	79.536	40	39.768	0.232
Iraq	7422.3	1660.6	77.627	39	38.814	0.186
Guyana	5375.5	1314.9	75.538	38	37.769	0.231
Botswana	405.1	104.3	74.248	38	37.124	0.876
Serbia	814.4	209.9	74.230	38	37.115	0.885
Bangladesh	8365.8	2214.0	73.535	37	36.768	0.232
Liechtenstein	243.7	66.0	72.903	37	36.452	0.548
Thailand	63328.2	17719.3	72.020	37	36.010	0.990
Bosnia and Herzegovina	179.1	53.0	70.412	36	35.206	0.794
China	438947.4	143545.7	67.298	34	33.649	0.351
North Macedonia	173.7	60.6	65.094	33	32.547	0.453
Indonesia	28084.7	10202.1	63.674	32	31.837	0.163
Taiwan	116264.0	42336.9	63.586	32	31.793	0.207
Angola	1869.2	682.4	63.496	32	31.748	0.252
Fiji	259.3	94.7	63.483	32	31.741	0.259
Libya	1465.6	567.2	61.297	31	30.648	0.352
Moldova	136.5	53.6	60.709	31	30.355	0.645
Switzerland	63425.3	24962.0	60.644	32	30.322	1.678
South Africa	14655.8	5819.0	60.296	31	30.148	0.852
Nauru	2.2	0.9	59.056	30	29.528	0.472
Algeria	2461.6	1014.5	58.787	30	29.393	0.607
Pakistan	5123.7	2135.1	58.328	30	29.164	0.836
Tunisia	1123.2	503.6	55.165	28	27.583	0.417
Kazakhstan	2330.7	1077.1	53.788	27	26.894	0.106
India	87416.4	41752.7	52.237	27	26.119	0.881
South Korea	131549.2	65541.8	50.177	26	25.088	0.912

Notes: Column 2 reports US imports from each country in 2024 in million US dollars. Column 3 reports US exports to each country in 2024 in million US dollars. Column 4 reports the bilateral US deficit with each country in 2024, computed as $\frac{\text{imports}-\text{exports}}{\text{imports}}$. Column 5 reports the total tariff announced on April 2. Column 6 reports the deficit component of the total tariff, calculated as half the bilateral US deficit. Column 7 reports the round-up component of the total tariff, calculated as the difference between the total tariff and the deficit component of it.

Table 1: List of trading partners with which the US has a bilateral trade deficit (Part 2)

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Country Name	US imports	US exports	US deficit	Total tariff	Deficit tariff component	Round-up tariff component
Malaysia	52534.8	27704.8	47.264	24	23.632	0.368
Brunei	238.8	127.2	46.739	24	23.370	0.630
Japan	148208.6	79740.8	46.197	24	23.098	0.902
Vanuatu	13.7	7.6	44.384	23	22.192	0.808
Namibia	275.2	160.5	41.682	21	20.841	0.159
Côte d'Ivoire	1014.7	596.6	41.206	21	20.603	0.397
Jordan	3364.9	2030.8	39.648	20	19.824	0.176
European Union	605760.4	370189.2	38.889	20	19.444	0.556
Nicaragua	4622.3	2941.0	36.374	19	18.187	0.813
Zimbabwe	67.8	43.8	35.455	18	17.728	0.272
Philippines	14177.6	9297.3	34.423	18	17.211	0.789
Malawi	41.1	26.9	34.378	18	17.189	0.811
Israel	22217.5	14792.0	33.422	17	16.711	0.289
Zambia	169.1	113.8	32.729	17	16.365	0.635
Mozambique	216.1	149.7	30.718	16	15.359	0.641
Norway	6583.3	4592.0	30.248	16	15.124	0.876
Venezuela	5987.9	4232.0	29.323	15	14.662	0.338
Nigeria	5699.2	4174.3	26.756	14	13.378	0.622
Chad	80.8	59.9	25.830	13	12.915	0.085
Equatorial Guinea	127.6	95.4	25.275	13	12.637	0.363
Cameroon	248.8	193.1	22.403	12	11.202	0.798
Dem. Congo	323.1	253.3	21.608	11	10.804	0.196
Receive baseline 10% tariff:						
New Zealand	5617.0	4499.4	19.897	10	9.948	0.052
Uganda	132.6	106.3	19.850	10	9.925	0.075
Ghana	1171.7	967.3	17.446	10	8.723	1.277
Costa Rica	11634.9	9676.8	16.830	10	8.415	1.585
Saint Helena	10.6	9.0	15.235	10	7.617	2.383
Papua New Guinea	79.2	67.2	15.098	10	7.549	2.451
New Caledonia	33.8	29.4	13.149	10	6.575	3.425
Ecuador	8524.3	7531.7	11.645	10	5.822	4.178
Trinidad and Tobago	3325.9	2939.7	11.612	10	5.806	4.194
Turkey	16745.7	15293.0	8.675	10	4.337	5.663
Iceland	1072.6	985.8	8.091	10	4.045	5.955
Mayotte	1.9	1.8	3.815	10	1.908	8.092
Gabon	171.7	171.1	0.351	10	0.175	9.825

Notes: Column 2 reports US imports from each country in 2024 in million US dollars. Column 3 reports US exports to each country in 2024 in million US dollars. Column 4 reports the bilateral US deficit with each country in 2024, computed as $\frac{\text{imports} - \text{exports}}{\text{imports}}$. Column 5 reports the total tariff announced on April 2. Column 6 reports the deficit component of the total tariff, calculated as half the bilateral US deficit. Column 7 reports the round-up component of the total tariff, calculated as the difference between the total tariff and the deficit component of it.

Table 2: List of countries in the Compustat Global sample

Country	Number of firms	Country	Number of firms
Bangladesh	352	Nigeria	144
Botswana	22	Norway	246
China	5132	Pakistan	445
Côte d'Ivoire	33	Philippines	268
European Union	3561	Serbia	27
Ecuador	8	South Africa	215
Ghana	23	South Korea	2453
Iceland	26	Sri Lanka	274
India	5060	Switzerland	224
Indonesia	913	Taiwan	2026
Israel	493	Thailand	905
Japan	4035	Trinidad and Tobago	22
Jordan	182	Tunisia	78
Kazakhstan	27	Turkey	586
Liechtenstein	2	Uganda	11
Malawi	14	Venezuela	23
Malaysia	1042	Vietnam	687
Mauritius	67	Zambia	20
New Zealand	114	Zimbabwe	4

Notes: This table reports the list of countries with listed firms in Compustat Global and the number of firms in each country (that do not have a missing industry classification).

Table 3: List of countries in the exchange rate sample

Angola	Laos	Serbia
Bangladesh	Madagascar	South Africa
Botswana	Malawi	South Korea
Cambodia	Malaysia	Sri Lanka
China	Mozambique	Switzerland
Costa Rica	Myanmar	Thailand
European Union	New Zealand	Trinidad and Tobago
Ghana	Nicaragua	Tunisia
Guyana	Nigeria	Turkey
Iceland	North Macedonia	Uganda
India	Norway	Venezuela
Israel	Papua New Guinea	Vietnam
Japan	Philippines	Zambia
Kazakhstan		

Notes: This table reports the list of countries in the exchange rate sample. The list of countries with a floating exchange rate regime are: Costa Rica, the European Union, Iceland, India, Israel, Japan, Kazakhstan, Madagascar, Malaysia, New Zealand, Norway, Philippines, South Africa, South Korea, Switzerland, Thailand, and Uganda. Note that the IMF's AREAER distinguishes between free floating and floating regimes, which we group. In our sample, only the European Union, Japan and Norway are classified as having a free floating regime.

The list of countries or territories in Table 1 excluded from the exchange rate sample because they do not have their own currency, are in a monetary union, or are pegged is: Cameroon (uses the Central African CFA franc, which is pegged to the euro), Chad (uses the Central African CFA franc), Côte d'Ivoire (uses the West African CFA franc, which is pegged to the euro), Ecuador (uses the US dollar), Equatorial Guinea (uses the Central African CFA franc), Falkland Islands (pegged to the British pound), Fiji (pegged to a basket of currencies), Gabon (uses the Central African CFA franc), Iraq (pegged to the US dollar), Jordan (pegged to the US dollar), Lesotho (pegged to the South African Rand), Libya (pegged to a basket of currencies), Liechtenstein (uses the Swiss franc), Mayotte (uses the euro), Namibia (pegged to the South African rand), Nauru (uses the Australian dollar), New Caledonia (uses the CFP franc), and Saint Helena (pegged to the British pound). The list of countries or territories in Table 1 excluded due to lack of available exchange rate data within the horizons we study is: Algeria, Bosnia and Herzegovina, Brunei, Dem. Rep. Congo, Indonesia, Mauritius, Moldova, Pakistan, Taiwan, and Vanuatu. Finally, we exclude Zimbabwe due to its hyperinflation and currency instability.

Table 4: List of tariffs for countries in exchange rate sample: Accounting for product exclusions

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Country name	Total tariff	Tariff accounting for exclusions	Exclusion component	Country name	Total tariff	Tariff accounting for exclusions	Exclusion component
Angola	32	5.598	-26.402	New Zealand	10	9.339	-0.661
Bangladesh	37	36.906	-0.094	Nicaragua	19	17.292	-1.708
Botswana	38	38.000	-0.000	Nigeria	14	1.027	-12.973
Cambodia	49	47.855	-1.145	North Macedonia	33	32.995	-0.005
China	34	32.362	-1.638	Norway	16	10.251	-5.749
Costa Rica	10	8.979	-1.021	Papua New Guinea	10	9.868	-0.132
European Union	20	13.811	-6.189	Philippines	18	16.454	-1.546
Ghana	10	3.113	-6.887	Serbia	38	36.090	-1.910
Guyana	38	2.055	-35.945	South Africa	31	19.808	-11.192
Iceland	10	7.188	-2.812	South Korea	26	23.213	-2.787
India	27	21.164	-5.836	Sri Lanka	44	43.923	-0.077
Israel	17	11.773	-5.227	Switzerland	32	19.987	-12.013
Japan	24	21.936	-2.064	Thailand	37	35.456	-1.544
Kazakhstan	27	4.254	-22.746	Trinidad and Tobago	10	4.848	-5.152
Laos	48	46.508	-1.492	Tunisia	28	27.456	-0.544
Madagascar	47	39.736	-7.264	Turkey	10	9.045	-0.955
Malawi	18	17.998	-0.002	Uganda	10	9.981	-0.019
Malaysia	24	19.115	-4.885	Venezuela	15	1.989	-13.011
Mozambique	16	12.209	-3.791	Vietnam	46	45.270	-0.730
Myanmar	45	44.817	-0.183	Zambia	17	14.306	-2.694

Notes: Columns 2 and 6 report the total tariff announced on April 2. Columns 3 and 7 report the tariff computed as a weighted average accounting for exclusions (see text). Columns 4 and 8 report the difference between the weighted tariff accounting for exclusions and the total tariff.

Table 5: The effect of tariffs on stock prices

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Total tariff	-0.031 (0.022)						
Round-up component		-0.234* (0.133)	-0.333** (0.129)	-0.258** (0.126)	-0.355*** (0.116)	-0.408** (0.172)	-0.306* (0.155)
Deficit component		-0.042* (0.024)	-0.047* (0.024)	-0.046** (0.021)	-0.051** (0.021)	-0.061** (0.024)	-0.045* (0.024)
Observations	29764	29764	8024	23895	5431	1448	3983

Notes: This table reports the results of the estimation of equation (3). Columns 1 and 2 include all firms in the sample. Column 3 excludes large countries (China, the European Union, India, Indonesia, Japan, South Korea, and Turkey) as defined in the text. Column 4 removes firms in industry-country combinations with product exclusions as explained in the main text. Column 5 omits both large countries and industry-country combinations with product exclusions. Column 6 further restricts the sample to firms in tradable industries. Column 7 restricts the sample to firms in nontradable industries. The dependent variable is $(100 \times)$ the log change in the stock price (in local currency) between the closing value of April 2 and April 3 2025. ***, **, and * indicate statistical significance at the 1%, 5% and 10% level.

Table 6: The effect of tariffs on stock prices: Robustness

	(1)	(2)	(3)	(4)	(5)
Round-up component	-0.329** (0.128)	-0.345*** (0.112)	-0.335*** (0.112)	-0.404*** (0.109)	-0.392*** (0.107)
Deficit component	-0.050** (0.024)	-0.051** (0.023)	-0.048** (0.021)	-0.049** (0.024)	-0.047** (0.022)
Observations	6576	5420	5431	5418	5259

Notes: This table reports the results of the estimation of equation (3). All columns exclude large countries (China, the European Union, India, Indonesia, Japan, South Korea, and Turkey) as defined in the text and remove industry-country combinations with product exclusions as explained in the main text. Column 1 does not include industry fixed effects. Column 2 includes NAICS 3-digit industry fixed effects. Column 3 uses winsorized price changes at the 1st and 99th percentiles. Column 4 is restricted to countries with at least 10 listed firms. Column 5 excludes Ecuador and South Africa as discussed in the text. The dependent variable is $(100 \times)$ the log change in the stock price (in local currency) between the closing value of April 2 and April 3 2025. ***, **, and * indicate statistical significance at the 1%, 5% and 10% level.

Table 7: The effect of tariffs on exchange rates

	(1)	(2)	(3)	(4)	(5)
Total tariff	0.0026 (0.0110)				
Round-up component		-0.0605 (0.1197)	-0.1693 (0.1289)	0.0280 (0.1436)	-0.0532 (0.0692)
Deficit component		0.0002 (0.0120)	-0.0032 (0.0110)	0.0089 (0.0122)	-0.0035 (0.0062)
Observations	38	38	33	23	22

Notes: This table reports the results of the estimation of equation (4). Columns 1 through 3 consider a 24-hour window; column 4 considers a 12-hour window and column 5 considers a 3 hour window. Column 3 excludes large countries (China, the European Union, India, Japan, South Korea, and Turkey) as defined in the text. The dependent variable is $(100 \times)$ the log change in the exchange rate. Exchange rates are defined in terms of local currency per US dollar. ***, **, and * indicate statistical significance at the 1%, 5% and 10% level.

Table 8: The effect of tariffs on exchange rates: Distinguishing exchange rate regimes

	(1)	(2)	(3)	(4)	(5)
Total tariff	-0.0038 (0.0136)				
Total tariff $\times$ Floating	0.0079 (0.0245)				
Floating	-0.6097 (0.6667)	-0.1603 (0.8190)	0.0955 (0.8005)	-0.2096 (0.9982)	0.3224 (0.4545)
Round-up component		0.0439 (0.1704)	0.0213 (0.2444)	0.0874 (0.1907)	-0.0167 (0.0847)
Deficit component		-0.0016 (0.0158)	-0.0019 (0.0150)	0.0099 (0.0185)	-0.0006 (0.0089)
Round-up component $\times$ Floating		-0.2506 (0.2392)	-0.2631 (0.2915)	-1.3217** (0.5961)	-0.4250* (0.2429)
Deficit component $\times$ Floating		-0.0000 (0.0262)	-0.0012 (0.0249)	0.0406 (0.0378)	-0.0060 (0.0136)
Observations	38	38	33	23	22

Notes: This table reports the results of the estimation of equation (5). Columns 1 through 3 consider a 24-hour window; column 4 considers a 12-hour window and column 5 considers a 3 hour window. Column 3 excludes large countries (China, the European Union, India, Japan, South Korea, and Turkey) as defined in the text. The dependent variable is $(100 \times)$ the log change in the exchange rate. Exchange rates are defined in terms of local currency per US dollar. ***, **, and * indicate statistical significance at the 1%, 5% and 10% level.

Table 9: The effect of tariffs on exchange rates: Accounting for excluded products

	(1)	(2)	(3)	(4)	(5)	(6)
Weighted tariff	0.0002 (0.0096)		-0.0059 (0.0108)		0.0037 (0.0134)	
Round-up component		-0.0561 (0.1225)		0.0719 (0.1792)		0.1359 (0.2397)
Deficit component		0.0003 (0.0122)		0.0004 (0.0164)		0.0154 (0.0247)
Exclusions component		-0.0046 (0.0184)		-0.0137 (0.0213)		-0.0278 (0.0769)
Weighted tariff $\times$ Floating			0.0158 (0.0246)		-0.0035 (0.0296)	
Floating			-0.7159 (0.5328)	0.0173 (0.8719)	-0.2190 (0.6784)	0.0310 (1.3104)
Round-up component $\times$ Floating				-0.2785 (0.2478)		-1.5158** (0.6664)
Deficit component $\times$ Floating				0.0034 (0.0285)		0.0497 (0.0469)
Exclusions component $\times$ Floating				0.0383 (0.0477)		0.0558 (0.0861)
Observations	38	38	38	38	23	23

Notes: This table reports the results of the estimation of equation (5). Columns 1 through 4 consider a 24-hour window; columns 5 and 6 consider a 12-hour window. The dependent variable is (100 $\times$) the log change in the exchange rate. Exchange rates are defined in terms of local currency per US dollar.

***, **, and * indicate statistical significance at the 1%, 5% and 10% level.

Table 10: The effect of tariffs on exchange rates: Including controls

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Round-up component	0.0698 (0.1746)	0.1286 (0.1821)	0.2210 (0.1596)	0.2669 (0.1808)	0.1021 (0.1814)	0.1087 (0.1665)	0.1305 (0.1696)	0.3113 (0.1836)
Deficit component	0.0007 (0.0160)	0.0173 (0.0207)	-0.0024 (0.0177)	0.0103 (0.0255)	-0.0028 (0.0167)	0.0076 (0.0154)	0.0047 (0.0159)	0.0183 (0.0314)
Exclusions component	-0.0123 (0.0207)	0.0278 (0.0385)	0.0070 (0.0251)	0.0372 (0.0477)	-0.0145 (0.0213)	-0.0092 (0.0198)	-0.0100 (0.0199)	0.0306 (0.0534)
Round-up comp. $\times$ Float.	-0.2822 (0.2415)	-0.2772 (0.2462)	-0.3108 (0.1990)	-0.3499 (0.2226)	-0.3256 (0.2517)	-0.3076 (0.2298)	-0.3421 (0.2349)	-0.4242* (0.2272)
Deficit comp. $\times$ Float.	0.0022 (0.0278)	0.0049 (0.0308)	0.0112 (0.0240)	0.0052 (0.0302)	0.0048 (0.0285)	-0.0177 (0.0279)	-0.0167 (0.0281)	-0.0231 (0.0364)
Exclusions comp. $\times$ Float.	0.0251 (0.0472)	0.0164 (0.0540)	-0.0040 (0.0389)	-0.0073 (0.0480)	0.0329 (0.0479)	0.0438 (0.0444)	0.0402 (0.0448)	-0.0038 (0.0499)
Floating	-0.0265 (0.8500)	0.0970 (1.0160)	-0.1705 (0.7866)	0.1639 (1.1094)	0.1710 (0.8835)	0.7362 (0.8804)	0.8719 (0.9006)	1.2888 (1.2896)
Observations	38	38	27	27	38	37	37	26

Notes: This table reports the results of the estimation of equation (5). All columns consider a 24-hour window. Columns 1 through 4 include controls capturing the trade structure of each country. Specifically, column 1 controls for the ratio of exports to the US over total exports. Column 2 controls for the share of exports to the US that are industrial supplies and by the share of exports to the US that are commodities. Column 3 controls for the share of dollar invoicing in total exports. Column 4 controls for all of the above. Columns 5 through 7 control for financial variables. Specifically, column 5 controls for the share of liabilities with the US over total liabilities in portfolio investment. Column 6 controls for the Chinn-Ito index of capital account openness. Column 7 controls for the share of liabilities with the US over total liabilities in portfolio investment and by the Chinn-Ito index. Finally, column 8 controls for all of the above (including trade structure controls and financial controls). The dependent variable is $(100 \times)$ the log change in the exchange rate. Exchange rates are defined in terms of local currency per US dollar. ***, **, and * indicate statistical significance at the 1%, 5% and 10% level.

A. APPENDIX

A.1. Links to official documents related to April 2 tariffs

- [January 20, 2025 Memorandum](#)
- [February 13, 2025 Memorandum](#)
- [April 2, 2025 Executive Order](#)
- [USTR Reciprocal Tariff Calculations](#)

A.2. Link to Census Bureau trade data

- [US Census Bureau: Trade in Goods by Country](#)

A.3. Additional data sources

- The share of exports to the US in total exports for each country is computed with data from COM-TRADE for 2023, obtained through the World Integrated Trade Solution (WITS).
- The share of each country's exports to the US that corresponds to intermediate inputs is computed using data on US imports by the Census Bureau disaggregated by HS 10-digit product and by country for 2024. HS 10-digit codes are matched to Census Bureau end-use categories. We consider the category "industrial supplies" as representing intermediate inputs.
- The share of each country's exports to the US that corresponds to commodities is computed using data on US imports by the Census Bureau disaggregated by HS 10-digit product and by country for 2024. We consider the following HS chapters as corresponding to commodities: 1 through 14 and 25 through 27.
- The share of each country's total exports invoiced in US dollars is computed using data from [Boz et al. \[2022\]](#). We use data from the most recent year available for each country.
- Data on portfolio investment is obtained from the International Monetary Fund (IMF)'s Portfolio Investment Positions by Counterpart Economy dataset (formerly Coordinated Portfolio Investment Survey, or CPIS). The data corresponds to year 2023.
- We use the Chinn-Ito index (for year 2022) as a measure of capital account openness [[Chinn and Ito, 2006](#)].

A.4. Appendix to Section 4: Model

We provide microfoundations to the model in Section 4. The key assumption is that firms produce with an increasing marginal cost, following Benguria and Saffie [2024] and Almunia et al. [2021].

We assume representative firms. Consider an exporting firm facing a demand in each market $m = (US, ROW)$ given by:

$$q_e = A_m \cdot p_e^{-\sigma}. \quad (10)$$

In this expression, $\sigma > 1$ is the elasticity of substitution, q_e is the quantity demanded and p_e is the price. The demand level is $A_m = E_m \cdot P_m^{\sigma-1}$ (which is a function of the price index P_m and aggregate expenditure E_m) and is treated as exogenous. The firm's variable cost is:

$$\frac{1}{\lambda + 1} \cdot (q_{e,US} + q_{e,ROW})^{\lambda+1}. \quad (11)$$

In this expression, $\lambda > 0$ represents the degree to which marginal cost is increasing.

Also note that tariffs (τ_{US} and τ_{ROW}) are represented as iceberg trade costs, such that $\tau_{US} > 1$ and $\tau_{ROW} > 1$.

The firm maximizes total profits in both markets (the sum of profits across markets). Due to the increasing marginal cost assumption, firms' optimal choices across markets are interdependent. In consequence, the export value of the firm in the US and the rest of the world (ROW) are:

$$r_{US} = \left(\frac{\sigma - 1}{\sigma} \right)^{\frac{\sigma-1}{\lambda\sigma+1}} \cdot (\lambda + 1)^{\frac{\sigma-1}{\lambda\sigma+1}} \cdot A_{US} \cdot \tau_{US}^{1-\sigma} \cdot \left(A_{US} \cdot \tau_{US}^{1-\sigma} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \right)^{\frac{-\lambda(\sigma-1)}{\lambda\sigma+1}}, \quad (12)$$

and

$$r_{ROW} = \left(\frac{\sigma - 1}{\sigma} \right)^{\frac{\sigma-1}{\lambda\sigma+1}} \cdot (\lambda + 1)^{\frac{\sigma-1}{\lambda\sigma+1}} \cdot A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \cdot \left(A_{US} \cdot \tau_{US}^{1-\sigma} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \right)^{\frac{-\lambda(\sigma-1)}{\lambda\sigma+1}}. \quad (13)$$

From these expressions, we find $\frac{dr_{US}}{d\tau_{US}} < 0$ and $\frac{dr_{ROW}}{d\tau_{US}} > 0$

Effect of a change in the US tariff on the exchange rate Here, we compute the derivative of the exchange rate e defined in equation (9) to a change in the US tariff τ_{US} .

$$\frac{de}{d\tau_{US}} = \frac{\frac{d((1-\phi)r_{ROW})}{d\tau_{US}} \cdot (r_{US} + \phi \cdot r_{ROW}) - \frac{d(r_{US} + \phi \cdot r_{ROW})}{d\tau_{US}} \cdot ((1-\phi) \cdot r_{ROW})}{(r_{US} + \phi r_{ROW})^2}. \quad (14)$$

The denominator in this expression is positive. The numerator can be simplified to:

$$\frac{dr_{US}}{d\tau_{US}} \cdot (-(1-\phi)r_{ROW}) + \frac{dr_{ROW}}{d\tau_{US}} \cdot ((1-\phi)r_{US}). \quad (15)$$

Given that $\frac{dr_{US}}{d\tau_{US}} < 0$ and $\frac{dr_{ROW}}{d\tau_{US}} > 0$ we have that $\frac{de}{d\tau_{US}} > 0$ meaning that a higher US tariff leads to an appreciation of the producer currency.

Endogenous invoice currencies We extend our analysis to consider a case in which the invoice currency choice changes in response to a US tariff. In particular, we assume that following a US tariff, exports to the US remain invoiced in dollars and exports to the rest of the world shift toward a higher share of invoicing in producer currency (i.e., invoicing moves away from the dollar). This means that the parameter ϕ is endogenous and $\frac{d\phi}{d\tau_{US}} < 0$.

We compute again the derivative of the exchange rate to a change in the US tariff considering this additional adjustment margin.

$$\frac{de}{d\tau_{US}} = \frac{\partial e}{\partial \tau_{US}} + \frac{\partial e}{\partial \phi} \cdot \frac{d\phi}{d\tau_{US}}. \quad (16)$$

The first term on the right, $\frac{\partial e}{\partial \tau_{US}}$, was calculated earlier and is positive. Below, we show that the second term on the right, $\frac{\partial e}{\partial \phi} \cdot \frac{d\phi}{d\tau_{US}}$, is also positive. We have that $\frac{d\phi}{d\tau_{US}}$ is negative by assumption, as stated earlier. Below, we see that $\frac{\partial e}{\partial \phi}$ is also negative, so the product is positive. We have that:

$$\frac{\partial e}{\partial \phi} = \frac{-r_{ROW} \cdot (r_{US} + r_{ROW})}{(r_{US} + \phi r_{ROW})^2}. \quad (17)$$

Thus, the appreciation of local currency in response to a US tariff is stronger when we consider endogenous currency choice along the lines discussed.

Effect of a change in the US tariff on profits and stock prices Our model also addresses the response of stock prices to the US tariff. We assume stock prices are a function of the present discounted value of future profits. Below, we compute the response of profits to the US tariff.

The total profits of the firm are:

$$\pi_{Total} = \pi_{US} + \pi_{ROW} = \left(\left(\frac{\sigma-1}{\sigma} \right)^{\frac{\sigma-1}{\lambda\sigma+1}} - \left(\frac{\sigma-1}{\sigma} \right)^{\frac{\sigma(\lambda+1)}{\lambda\sigma+1}} \right) \cdot (\lambda+1)^{\frac{\sigma-1}{\lambda\sigma+1}} \cdot \left(A_{US} \cdot \tau_{US}^{1-\sigma} + A_{ROW} \cdot \tau_{ROW}^{1-\sigma} \right)^{\frac{\lambda+1}{\lambda\sigma+1}}. \quad (18)$$

From this, we see that $\frac{d\pi_{Total}}{d\tau_{US}} < 0$ given that $\sigma > 1$.